Facility Sustainability Review: Canteen and Dining Hall Operations

Document ID: DOC_BLD_012 | **Type:** FACILITY_REPORT | **Publication:** 2024-08-28 | **Institution:** VESIT (VESIT Green Campus, Chembur)

FACILITY SUSTAINABILITY REVIEW: CANTEEN AND DINING HALL OPERATIONS

Document Reference: VESIT/FAC/CANTEEN/2024/01

Facility Code: BLD_CAN (Canteen Building)

Jurisdiction: Canteen Oversight Committee and Campus Facilities (DEP_ADMN).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. OPERATIONAL SCOPE AND RESOURCE METRICS

The Canteen building (BLD_CAN) occupies 2,100 m2 of built infrastructure comprising an industrial commercial kitchen, two student dining halls, a faculty dining lounge, and outdoor shaded eating patios. Serving a campus community of 3,200 individuals, the facility processes over 2,200 meals, snacks, and beverages every academic working day.

Because of intensive food preparation, utensil sanitation, and catering operations, BLD_CAN represents the campus's primary hub of solid organic waste generation, high daily water consumption, and commercial cooking fuel use:

- Annual Electricity Consumption: 192,000 kWh (specific intensity: 91.4 kWh/m2/year), driven by commercial refrigeration, induction warmers, ventilation exhaust hoods, and drinking water chillers.

- Annual Municipal Water Consumption: 13,440,000 litres (28.0% of total campus freshwater intake).

- Annual Solid Organic Waste: 28,728 kg of wet vegetable peelings, uneaten food scraps, and coffee grounds.

2. ON-SITE ANAEROBIC BIOMETHANATION PLANT (INIT_BIOGAS_CANTEEN)

To address wet food waste, the institution commissioned project INIT_BIOGAS_CANTEEN directly adjacent to the kitchen loading dock. The facility houses a 250 kg/day prefabricated anaerobic biomethanation reactor utilizing a microbial consortia optimized for starch, cellulose, and lipid degradation.

During 2024:

- Ingested 19,200 kg of source-segregated canteen kitchen scraps.

- Generated 4,380 m3 of methane-rich biogas (average 12.0 m3/day).

- Replaced 168 commercial LPG cylinders (19 kg each) annually, piped directly to two kitchen burners.

- Delivered direct fuel cost savings of INR 3,00,000 per year while abating 48,000 kg CO2e of fugitive methane and fossil fuel combustion emissions.

- Produced 1,400 kg of enriched liquid bioslurry and solid bio-manure utilized on campus botanical grounds.

3. SINGLE-USE PLASTIC ELIMINATION AND WASTE PACKAGING

In compliance with the Maharashtra Plastic and Thermocol Products notification and campus Circular Economy guidelines (DOC_WST_018):

- 100% Transition to Reusable Tableware: The canteen completely eliminated single-use plastic plates, bowls, cups, and polystyrene foam trays, transitioning to high-grade 304 food-grade stainless steel thali trays, copper-bottom water jugs, and ceramic cups.

- Elimination of Packaged Water Bottles: Replaced the retail sale of 500ml single-use PET bottled water with three high-capacity chilled commercial reverse osmosis (RO) bottle refill stations. This intervention eliminates over 42,000 disposable plastic bottles annually.

- Takeaway Packaging Protocol: Banned plastic takeaway wraps; all takeaway food containers are mandated to use certified biodegradable sugarcane bagasse clamshells and birchwood cutlery.

4. WATER RECOVERY AND GREYWATER CYCLING

Commercial RO purification currently discharges 14,000 litres/day of mineral-rich reject water. A plumbing retrofit currently underway routes this reject water to the dual-flush toilet tanks of BLD_CAN and the adjacent Academic Block A (BLD_ACA). Concurrently, a 15,000 L/day constructed wetland reed-bed system is scheduled for 2025 behind BLD_CAN to biologically purify greasy kitchen greywater for landscape irrigation.

These integrated canteen operations form the backbone of institutional compliance with targets TRG_ZERO_WASTE and TRG_WATER_NEUTRAL, directly demonstrating compliance with United Nations SDG 6, SDG 12, and SDG 13.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_BLD_012**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*